

DOMANDA

Dati i segnali

$$x(t) \leftrightarrow X(f)$$

$$y(t) \leftrightarrow Y(f)$$

$$X(f) = \int_{-\infty}^{\infty} dt x(t) e^{-j2\pi ft}$$

$$Y(f) = \int_{-\infty}^{\infty} dt y(t) e^{-j2\pi ft}$$

allora vale $\int_{-\infty}^{\infty} dt x(t) y^*(t) = \int_{-\infty}^{\infty} df X(f) Y^*(f)$

se $x(t) = y(t)$ vale

$$\int_{-\infty}^{\infty} dt |x(t)|^2 = \int_{-\infty}^{\infty} df |X(f)|^2$$

DIMOSTRAZIONE:

il prodotto nel dominio del tempo è la convoluzione nel dominio della frequenza

$$x(t) y^*(t) \leftrightarrow \{X(f)\}^* \{Y^*(-f)\}$$

$$\Rightarrow \{X(f)\}^* \{Y^*(-f)\} \stackrel{\Delta}{=} \int_{-\infty}^{\infty} d\lambda X(\lambda) Y^*(\lambda - f)$$

per la proprietà di simmetria degli FT $Y^*(f - \lambda) = Y^*(\lambda - f)$

siccome l'area di un segnale e' la sua trasformata di Fourier valutata in 0

$$\int_{-\infty}^{\infty} dt x(t) y(t) = \int_{-\infty}^{\infty} d\lambda X(\lambda) Y^*(\lambda - f) \Big|_{f=0} = \int_{-\infty}^{\infty} d\lambda X(\lambda) Y^*(\lambda)$$

□

$$\int_{-\infty}^{+\infty} \text{sinc}^2(t) \text{sinc}(-2t-1) dt =$$

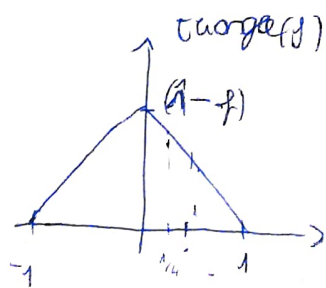
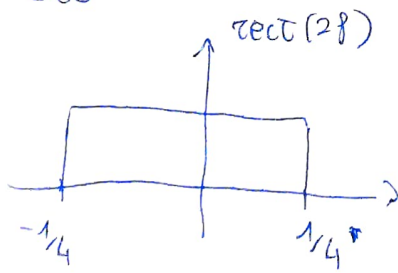
considero $x(t) = \text{sinc}^2(t)$ $y(t) = \text{sinc}(-2t-1) = \text{sinc}(t)$, oppure la trasformata di Fourier

$$X(f) = \text{triangle}(f), \quad y(t) = \text{sinc}\left(\frac{-t+1/2}{1/2}\right)$$

$$Y(f) = 2 \text{rect}(2f) e^{j2\pi \frac{1}{2} f}$$

~~scrittura~~

$$\int_{-\infty}^{+\infty} df \text{triangle}(f) 2 \text{rect}(2f) e^{j2\pi \frac{1}{2} f}$$



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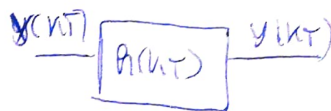


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$$= 2 \int_0^{1/4} 2f(1-f) = 2 \left[f - \frac{f^2}{2} \right]_0^{1/4} = 2 \left[\frac{1}{4} - \frac{1}{32} \right] = \frac{7}{16}$$

$$\textcircled{1} h(kT) = \frac{1}{T} 0,3^{-k} 10(-kT)$$

$$x(kT) = 1 + \sin(2\pi f_0 kT) \quad , \quad f_0 = \frac{1}{10T}$$



$$x(kT) = 1 + \sin(2\pi f_0 kT) = 1 + \cos(2\pi f_0 kT - \frac{\pi}{2}) = x_1(kT) + x_2(kT)$$

Siccome gli esponenziali complessi sono autofunzioni dei filtri

$$y(kT) = x_1(kT) H(f_{01}) + \cancel{x_2(kT)} |H(f_{02})| x_2(kT + \angle H(f_{02}))$$

$$H(f) = ?$$

$$\begin{aligned} H(f) &= T \sum_{k=-\infty}^0 \frac{1}{T} 0,3^{-k} e^{-j2\pi f kT} \\ &= \sum_{k=0}^{+\infty} 0,3^k e^{j2\pi f kT} = \sum_{k=0}^{+\infty} (0,3 e^{j2\pi f T})^k \\ &= \frac{1}{1 - 0,3 e^{j2\pi f T}} \end{aligned}$$

$$H(f_{01}) = H(0) = \frac{1}{1 - 0,3} = \frac{10}{7}$$

$$H(f_{02}) = H\left(\frac{1}{10T}\right) = \frac{1}{1 - 0,3 e^{j2\pi \frac{1}{10T} T}} = \frac{1}{1 - 0,3 e^{j2\pi \frac{1}{5}}}$$

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$$\textcircled{2} x(t) = \sum_{k=-\infty}^{+\infty} e^{-2t+6k} \quad 1(2t-6k) = \sum_{k=-\infty}^{+\infty} e^{-(2t-6k)} \quad 1(2t-6k)$$

$x(t)$ è periodico de periodizzazione

$$x_1(t) = e^{-t} \quad 1(t)$$

$$\tilde{x}_1(nT) = \sum_{k=-\infty}^{+\infty} e^{-(t-\frac{6}{2}k)} \quad 1(-t-\frac{6}{2}k)$$

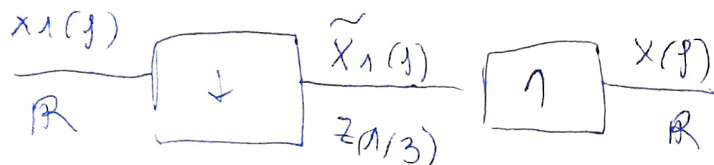
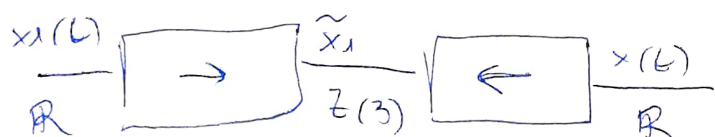
$$X_1(f) = \int_0^{+\infty} e^{-t} e^{-j2\pi f t} dt$$

$$= \int_0^{+\infty} e^{-t(1-j2\pi f)} dt = \left[\frac{e^{-t(1-j2\pi f)}}{-1-j2\pi f} \right]_0^{+\infty} = \frac{1}{1+j2\pi f}$$

$$\tilde{X}_1(f) = \frac{1}{1+j2\pi f \frac{1}{3}}$$

$$X(f) = 3 \sum_{k=-\infty}^{+\infty} \tilde{X}_1(f) \delta_R(t-n3)$$

$$= 3 \sum_n \frac{1}{1+j2\pi f \frac{1}{3}} \delta_R(t-n3)$$



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$$1. \dots \dots \dots = -12$$

③ $w(nT)$

$$R_w(f) = R_0(1 + \cos(2\pi fT))$$

$$x(nT) = w(nT) + w(nT - T) + w(nT - 2T)$$

$$h(nT) = \delta_z(t)(nT) + \delta_z(t)(nT - T) + \delta_z(t)(nT - 2T)$$

$$R_x = R_w |H(f)|^2$$

$$H(f) = 1 + e^{-j2\pi fT} + e^{-j2\pi f2T} = e^{-j2\pi fT} \left\{ 1 + e^{j2\pi fT} + e^{-j2\pi fT} \right\}$$

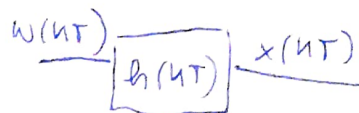
$$= e^{-j2\pi fT} (1 + 2\cos 2\pi fT)$$

$$= \cancel{e^{-j2\pi fT} (1 + \cos 2\pi fT)}$$

$$= \cancel{e^{-j2\pi fT} (\sin^2 \pi fT + \cos^2 \pi fT - \cos^2 \pi fT)} =$$

$$= e^{-j2\pi fT} (1 + 4\cos^2 \pi fT - 1) = e^{-j2\pi fT} (4\cos^2 \pi fT)$$

$2\cos^2 x = 1 + \cos(2x)$
 ~~$\cos^2 x = \frac{1 + \cos(2x)}{2}$~~
 $2\cos(2x) = \sqrt{2\cos^2 x - 1}^2$



$$R_x = R_0(1 + \cos(2\pi fT)) |H(f)|^2$$

$$M_x = \int R_x(f) = R_0 \int (1 + \cos(2\pi fT)) |H(f)|^2$$

④ $B = 20 \text{ kHz}$
 $M_T = 200 \text{ V}^2$
 $A_T = 2 \text{ dB/km}$
 $L = 60 \text{ km}$
 $R_0 = 10^{-18} \text{ V}^2/\text{Hz}$

$$\Delta = \frac{M_T A_T^2 m}{2 R_0 B}$$

$$-10 \log_{10} A_T^2 m = A_T \cdot L$$

$$-10 \log_{10} A_T^2 m = 2 \cdot 60 = 120$$

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$$\log_{10} A m^2 = -12$$

$$A m^2 = 10^{-12}$$

$$\Delta = \frac{200 \cdot 10^{-12}}{2 \cdot 10^{-18} \cdot 20K} = 5000$$

$$(5) g(t) = v_0 \operatorname{sinc}\left(\frac{2t}{T}\right)$$

$$h(t) = h_0 \operatorname{sinc}^2\left(\frac{t}{T}\right)$$

$$l(t) = l_0 \operatorname{sinc}\left(\frac{6t}{T}\right) \operatorname{sinc}\left(\frac{8t}{T}\right)$$

$$P_e = 2 \frac{3-2}{2} Q\left(\frac{v_d^2}{20n}\right)$$

$$= 2 Q\left(\frac{v_0^2}{20n}\right)$$

$$G(f) = v_0 \frac{T}{2} \operatorname{rect}\left(f \frac{T}{2}\right)$$

$$H(f) = h_0 T \operatorname{triang}\left(fT\right)$$

$$L(f) = \left\{ l_0 \frac{T}{8} \operatorname{rect}\left(f \frac{T}{8}\right) \right\}^* \left\{ \frac{T}{6} \operatorname{rect}\left(f \frac{T}{6}\right) \right\}$$

$$\frac{1}{2} \text{ Base maggiore} = \left(\frac{8}{T} + \frac{6}{T}\right) \frac{1}{2} = \frac{7}{T}$$

$$\frac{1}{2} \text{ Base minore} = \left(\frac{8}{T} - \frac{6}{T}\right) \frac{1}{2} = \frac{1}{T}$$

$$\text{altezza} = l_0 \frac{T}{8} \frac{T}{6} \frac{6}{T} = l_0 \frac{T}{8}$$

$$C(f) = L(f)H(f)G(f)$$

$$= v_0 \frac{T}{2} h_0 T l_0 \frac{T}{8} \operatorname{triang}\left(fT\right) \Rightarrow c(t) = v_0 \frac{T}{2} h_0 T l_0 \frac{T}{8} \frac{1}{T} \operatorname{sinc}^2\left(\frac{t}{T}\right)$$

$$C(0) = \frac{v_0}{16} h_0 l_0 T^2 = v_0$$

e di Nyquist.

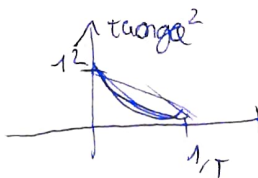
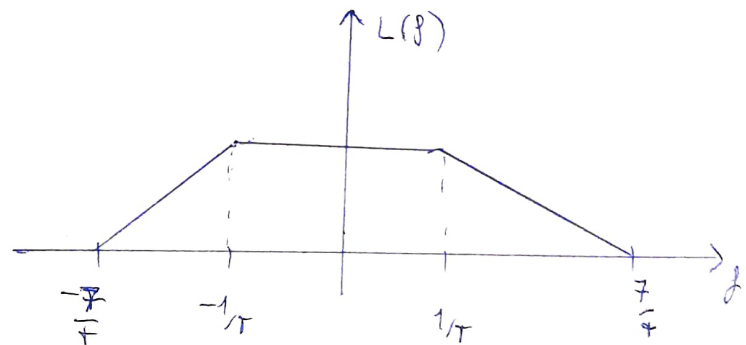
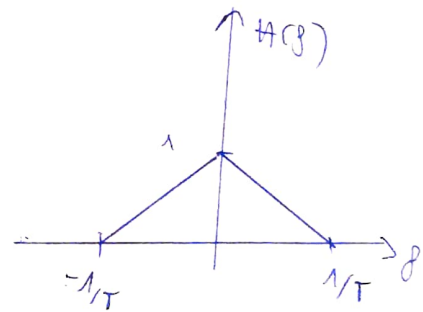
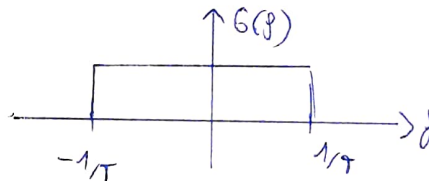
$$R_n = R_0 E_R \Rightarrow R_0 \int df |H(f)|^2$$

$$M_n = \sigma^2 n \text{ perché } m_n = 0$$

$$\Rightarrow \sigma^2 n = R_0 h_0^2 T^2 \int df \operatorname{triang}^2(fT)$$

$$= R_0 h_0^2 T^2 \frac{1}{T} \frac{2}{3} = R_0 h_0^2 T \frac{2}{3}$$

$$P_e = 2 Q\left(\frac{\frac{v_0^2}{16^2} h_0^2 l_0^2 T^4}{R_0 h_0^2 T \frac{2}{3}}\right)$$



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